Latent Reconciliation Histories, Missing Information, and Coordinate Changes in GPURec

A mathematical synthesis of two optimisation-agent transcripts

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Abstract

This report explains what two agents learned about gene-wise optimisation in GPURec's duplication–transfer–loss reconciliation model. The three base-2 log rates are already the natural parameters of the complete-data model, whose negative log-likelihood is convex. The observed objective can nevertheless be non-convex because reconciliation histories are latent: its curvature is complete-data multinomial information minus a posterior covariance of event counts, with an additional contribution from conditioning on family survival. Nonlinear coordinates such as square-root rates substantially improve the measured local geometry, but they do not reduce the production cost when curvature is carried by BFGS updates. The measured cost remains about eleven clade-weighted population passes. The EM-focused agent derived an EM iteration from the same structure, but its transcript ended before count extraction, missing-information fractions, or EM convergence were measured; those quantities therefore remain open rather than being inferred from the coordinate experiment.

1 Model and notation

For one gene family, let

$$\theta = (\theta_D, \theta_L, \theta_T) = (\log_2 D, \log_2 L, \log_2 T)$$

denote the duplication, loss, and transfer log rates. Speciation is the reference category with fixed logit zero. With

$$Z(\theta) = 1 + 2^{\theta_D} + 2^{\theta_L} + 2^{\theta_T},$$

the four event probabilities are

$$p_S = \frac{1}{Z}, \qquad p_k = \frac{2^{\theta_k}}{Z}, \quad k \in \{D, L, T\}. \tag{1}$$

The transfer probability is subsequently distributed among permissible recipient species by a factor that does not depend on the three rates. The fitted rates are constrained to $10^{-6} \le D, L, T \le 2$, or approximately $-19.93 \le \theta_k \le 1$.

A *complete history* h records every event of every lineage, including extinct lineages. Write $n_k(h)$ for its number of events of type k and $n_\bullet(h) = \sum_{k\in\{S,D,L,T\}} n_k(h)$. Rate-independent factors, including gene-tree split weights and recipient choices, are collected into w_h. Thus

$$P(h \mid \theta) = w_h \prod_{k\in\{S,D,L,T\}} p_k(\theta)^{n_k(h)}. \tag{2}$$

For observed gene-tree data y, let $\mathcal{H}_y$ be the compatible histories and define the unconditioned numerator

$$A_y(\theta) = \sum_{h \in \mathcal{H}_y} P(h \mid \theta).$$

If $q(\theta) = P_\theta(\text{family survives})$, GPURec uses the conditional likelihood

$$L_y(\theta) = \frac{A_y(\theta)}{q(\theta)}, \qquad \ell(\theta) = -\log_2 A_y(\theta) + \log_2 q(\theta). \tag{3}$$

The survival probability is obtained from extinction probabilities computed by a fixed point.

2 Why the log rates are the natural coordinates

For a fixed complete history, substitution of (1) into (2) gives

$$-\log_2 P(h \mid \theta) = n_\bullet(h) \log_2 Z(\theta) - \sum_{k \in \{D,L,T\}} n_k(h)\theta_k - \log_2 w_h. \tag{4}$$

This is a log-sum-exp minus a linear term. Consequently the complete-data negative log-likelihood is convex. More precisely, for

$$p = (p_D, p_L, p_T)^\mathsf{T}, \qquad C(p) = \operatorname{diag}(p) - pp^\mathsf{T},$$

its Hessian in base-2 log-rate coordinates is

$$\nabla^2_\theta[-\log_2 P(h \mid \theta)] = n_\bullet(h) \ln(2)\, C(p) \succeq 0. \tag{5}$$

The usual natural parameters are $\eta_k = \ln(D_k) = \ln(2)\theta_k$; multiplication by the common positive constant $\ln(2)$ is immaterial. Hence the existing log rates are already the natural coordinates, and the event counts are sufficient statistics. A nonlinear power transform may be computationally useful, but it is not more intrinsic to the complete-data statistical model.

3 Where observed-data non-convexity comes from

Marginalisation over histories changes the curvature. Define the complete-history score statistic

$$s(h) = \begin{pmatrix} n_D(h) \\ n_L(h) \\ n_T(h) \end{pmatrix} - n_\bullet(h)p.$$

For the unconditioned data term, the missing-information identity gives

$$H_y = \ln(2) \left\{ \mathbb{E}_y[n_\bullet] C(p) - \operatorname{Cov}_y[s(h)] \right\}. \tag{6}$$

Here the expectation and covariance are under the posterior distribution of histories conditional on y at the current parameter. The first term is the expected complete-data information. It is positive semidefinite and depends only on one expected total count and the four event probabilities. The second term is the information missing because the history was not observed. It is a posterior covariance and is also positive semidefinite, but it is subtracted. If alternative histories have very different event counts, this covariance can dominate the multinomial term and make H_y indefinite.

Survival conditioning in (3) must also be included. One direct form is

$$H = \ln(2)\left[\left\{\mathbb{E}_y[n_\bullet]C - \mathrm{Cov}_y(s)\right\} - \left\{\mathbb{E}_{\mathrm{surv}}[n_\bullet]C - \mathrm{Cov}_{\mathrm{surv}}(s)\right\}\right]. \tag{7}$$

Thus the survival histories enter with the opposite sign. The EM transcript also identified a useful equivalent augmentation. If $e = 1 - q$ is the extinction probability, then

$$\frac{1}{q} = \frac{1}{1-e} = \sum_{m=0}^{\infty} e^m.$$

The conditional likelihood may therefore be augmented by an observed family together with a geometrically distributed number of unobserved extinct "ghost" families. In this augmentation all expected event counts are positive and an identity of the form $H = I_c - I_m$ again holds with $I_c \succeq 0$ and $I_m \succeq 0$. The direct survival-difference counts and ghost-augmented counts give the same score: their discrepancy is proportional to the probability vector and is annihilated by the softmax Jacobian. The decomposition into complete and missing information depends on the chosen augmentation, whereas the observed Hessian H does not.

3.1 Why a coordinate map cannot remove the statistical ambiguity

Let ϕ be any smooth invertible coordinate system, write $\theta = t(\phi)$, let $J = \partial\theta/\partial\phi$, and put $g = \nabla_\theta \ell$. The transformed Hessian is

$$\nabla^2_\phi \ell = J^\mathsf{T} H J + sum_{k\in\{D,L,T\}} g_k \nabla^2_\phi t_k. \tag{8}$$

At a stationary point $g = 0$, only the congruence $J^\mathsf{T} H J$ remains. Sylvester's law of inertia then preserves the numbers of positive and negative eigenvalues. Away from a stationary point, a nonlinear map can add helpful curvature through the second term in (8); it cannot eliminate the posterior competition itself, and that added curvature vanishes at the optimum.

For example, power coordinates $\phi_k = r_k^\alpha = 2^{\alpha\theta_k}$ have inverse

$$\theta_k = \frac{\ln\phi_k}{\alpha\ln 2}, \qquad \frac{d\theta_k}{d\phi_k} = \frac{1}{\alpha\ln 2\,\phi_k}, \qquad \frac{d^2\theta_k}{d\phi_k^2} = -\frac{1}{\alpha\ln 2\,\phi_k^2}.$$

This explains both observations in the experiment: square-root rates can add useful curvature far from the optimum, while rate coordinates become numerically delicate near the 10^{-6} floor because the second derivative of the inverse map is very large.

4 The principled latent-variable iteration

The expectation–maximisation (EM) algorithm follows directly from (4). Its E-step computes posterior expected event counts

$$N_k = \mathbb{E}[n_k \mid y, \theta^{(r)}], \qquad N_\bullet = \sum_k N_k.$$

Using the positive ghost-family augmentation for survival conditioning, the M-step minimises the expected complete-data NLL and has the multinomial closed form

$$p_k^{(r+1)} = \frac{N_k}{N_\bullet}, \qquad \theta_k^{(r+1)} = \log_2\frac{N_k}{N_S}, \quad k \in \{D, L, T\}. \tag{9}$$

If instead one uses data-side minus survival-side counts directly, the same formula applies when all effective counts are positive. A nonpositive effective count sends the unconstrained optimum to a boundary; an active-set M-step or a Newton minimisation of the convex surrogate is then the appropriate implementation.

The box-constrained M-step also remains analytic. Suppose a set B of rates is pinned at log-rate bounds b_k, and let F denote the free rates. With

$$c = 1 + \sum_{k \in B} 2^{b_k},$$

the free solution is

$$\theta_k^{(r+1)} = \log_2 \left(\frac{N_k c}{N_S + \sum_{j \in B} N_j} \right), \qquad k \in F, \tag{10}$$

followed by the usual active-set and Karush–Kuhn–Tucker sign checks.

The existing reverse pass already accumulates adjoints with respect to the four free event log-probabilities before folding them through the softmax. If $a_k = \partial \ell / \partial \log_2 p_k = -N_k$, then the ordinary log-rate gradient is

$$g_k = a_k - p_k \sum_j a_j = p_k N_\bullet - N_k. \tag{11}$$

Thus the E-step should require essentially the same backward computation as a gradient pass.

Near an optimum, define the complete information I_c, observed information $I_o = H$, and missing information $I_m = I_c - I_o$. The local EM error-propagation matrix is

$$R_{\mathrm{EM}} = I_c^{-1} I_m, \tag{12}$$

up to a similar symmetric representation. Its spectral radius is the fraction of missing information and determines the linear convergence rate. A generalized eigenvalue larger than one is equivalent to negative observed curvature in the corresponding direction. This supplies a statistical explanation for why BFGS–Newton can be viewed as an acceleration: it attempts to learn $I_o = I_c - I_m$, whereas plain EM uses the inexpensive I_c.

Evidence boundary. The EM transcript stopped immediately after writing this derivation. It did not create the planned count-extraction script, validate (11), measure the missing-information fractions, or run EM, SQUAREM, or a hybrid method. The report therefore makes no numerical claim about EM's actual convergence on the Coleman data.

5 What the coordinate experiment measured

The completed experiment used the first 500 Coleman gene families and exact 3×3 Hessians at the common start, after three Adam steps, and at the fitted optimum. A distance is the largest of the three absolute log-rate gaps from the optimum, followed by the median across families. The post-Adam median distance was 2.48 log-rate units and the 90th percentile journey was about 8.5.

5.1 Direct evidence for competing reconciliation explanations

At the post-Adam point, duplication was above its eventual optimum in 72% of families and transfer was below its optimum in 93%. The median transfer gap was 2.09 log-rate units, or approximately a fourfold rate increase. Nevertheless, the local duplication gradient requested an *increase* in 69% of families. At artificially low transfer, duplication temporarily explains discordant gene histories; as transfer rises, those duplications are no longer needed.

This competition also appears in exact negative-curvature eigenvectors. At the post-Adam point, $320/500 = 64\%$ of Hessians were indefinite. The most negative direction had mean squared weights 83% duplication, 16% transfer, and 1% loss; its duplication and transfer components had opposite signs in 95% of indefinite families. At the common start, $410/500 = 82\%$ were indefinite and the corresponding direction was 63% loss, 31% transfer, and 6% duplication. These measurements are exactly the pattern predicted by the covariance term in (6): histories exchange duplication or loss events for transfer events.

5.2 Coordinate screening and exact-Hessian trajectories

Table 1 reports the principal coordinate results. "Indefinite" gives the percentage at the start, post-Adam point, and optimum. "Drift" is the median relative Frobenius distance between the post-Adam and optimum Hessians in the stated coordinates. Multi-step entries are GPU-evaluated, accepted exact-Hessian steps; one-step-only entries are from the CPU screening.

Table 1: Geometry and median distance to the optimum for candidate coordinates.

Coordinates	Indefinite start/Adam/opt.	Drift	Step 1	Step 2	Step 3	NLL excess after step 3
Log_2 rates, production radius rule	82/64/0.2%	1.01	1.65	1.40	0.75	697 bits
Log_2 rates, radius fixed at 2	82/64/0.2%	1.01	1.65	0.98	0.47	876 bits
Square-root rates	26/10/0%	0.90	1.27	0.55	0.07	141 bits
Cube-root rates	47/23/0.2%	0.56	1.30	0.61	0.08	46 bits
Square-root event probabilities	47/10/0.2%	0.86	1.40	0.49	0.04	55 bits
Linear rates	35/33/36%	3.70	1.73	—	—	—
Event-probability simplex	16/12/11%	3.02	1.77	—	—	—
Per-event log odds	86/53/0.2%	0.94	1.70	—	—	—
Log total rate plus event shares	30/7/6%	4.70	1.98	—	—	—
Oracle whitening by optimum Hessian	82/64/0.2%	2.33	2.57	—	—	—

Square-root rates reduced post-Adam indefiniteness from 64% to 10% and reduced the median optimum condition number from about 80 to 5.4; square-root probabilities achieved a condition number near 6.1. Cube-root rates gave the lowest Hessian drift, 0.56, and the smallest measured quadratic-model error in the screening. Exact-Hessian trajectories consequently improved from

$$2.48 \to 1.65 \to 0.98 \to 0.47 \quad \text{(fixed-radius log rates)}$$

to

$$2.48 \to 1.27 \to 0.55 \to 0.07 \quad \text{(square-root rates).}$$

The apparent indefiniteness of linear-rate and raw-probability coordinates at the constrained optimum is not evidence of an interior saddle. Many points lie at a rate bound, so the residual gradient term in (8) need not vanish; the inverse map also has derivatives of order r^{-2} near the floor.

The production trust radius itself imposes a lower bound on one-step progress. A perfect straight-line displacement toward the optimum, capped at two log-rate units, would still leave a median distance of 0.64. Moreover, after three exact steps only 5–11% of families in any coordinate system passed the actual certificate $\|g_{\text{projected}}\|_\infty < 10^{-3}$. Median geometric proximity is therefore not the same as convergence of the costly tail, particularly because large-clade families converge last.

5.3 Why improved geometry did not improve production cost

An exact Hessian costs about 7.7 whole-population gradient passes. Three exact-Hessian steps in square-root coordinates therefore cost approximately $3(7.7 + 1) \simeq 26$ passes. Taking one exact Hessian and then carrying it with BFGS cost 15.8 clade-weighted passes, compared with about 11.0 for the production replay.

The decisive production-like test used one gradient pass per round and a BFGS-carried curvature. Table 2 shows the state after twelve rounds. The native seeds were constructed in each coordinate system from the same start-to-post-Adam gradient pair.

Table 2: Production-like BFGS replay after twelve rounds.

Run	Frozen families	Live clades	Weighted cost	NLL excess
Log_2 rates, production seed	63%	21%	11.0	1.7 bits
Log_2 rates, native seed	58%	32%	11.01	2.0 bits
Square-root rates, native seed	37%	73%	11.76	22.7 bits
Cube-root rates, native seed	73%	24%	11.25	0.9 bits
Square-root probabilities, native seed	80%	19%	11.10	34.3 bits
Square-root rates, exact seed	80%	6%	15.8	—

Thus the useful nonlinear maps changed curvature but not end-to-end cost under the curvature scheme that production can afford. A rank-two BFGS update still has to learn a drifting 3×3 matrix from successive gradient pairs; the trust radius limits long moves; and the clade-weighted tail dominates wall time. The non-convexity is also structural rather than a mere scaling defect: the posterior over histories changes as transfer rises, so the Hessian changes with it.

6 Conclusions

The combined mathematical and empirical conclusions are:

1. The base-2 log rates are the model's natural coordinates and make every fixed-history complete-data objective convex.

2. Marginalising reconciliation histories subtracts a posterior score covariance from the multinomial information. Survival conditioning contributes an analogous term. This missing information is the source of the measured duplication–transfer and loss–transfer negative directions.

3. Nonlinear power coordinates can improve off-optimum curvature through a gradient-dependent Hessian term. Square-root and cube-root variants markedly improve exact-Newton trajectories, but no smooth map removes the latent ambiguity, and at a true stationary point coordinate changes preserve inertia.

4. With production's BFGS-carried curvature, the tested maps cost about 11.0–11.8 clade-weighted passes, while an exact-Hessian seed is too expensive. Reparameterisation therefore does not cut the Newton phase from roughly eleven passes to a few.
5. EM is the principled remaining experiment: its M-step is analytic and its E-step should be obtainable from the existing adjoints, but neither its counts nor its missing-information fraction nor its convergence trace was measured in the supplied transcript.

One-sentence verdict. The agents learned that the optimisation difficulty is caused primarily by posterior uncertainty between competing reconciliation histories, not by a poor choice of rate coordinates: nonlinear coordinates improve exact local geometry, but they do not reduce affordable production passes, and the proposed EM alternative remains theoretically motivated but experimentally untested.

Provenance

This synthesis is based on the two supplied JSONL task outputs. The completed reparameterisation study reported that it changed no repository source files and stored its scripts and measurements under the scratchpad's `reparam/` directory. The EM task wrote only its derivation record under `em/PROGRESS.md` before its transcript ended. This document is the only new source artifact created for the present request.

References

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